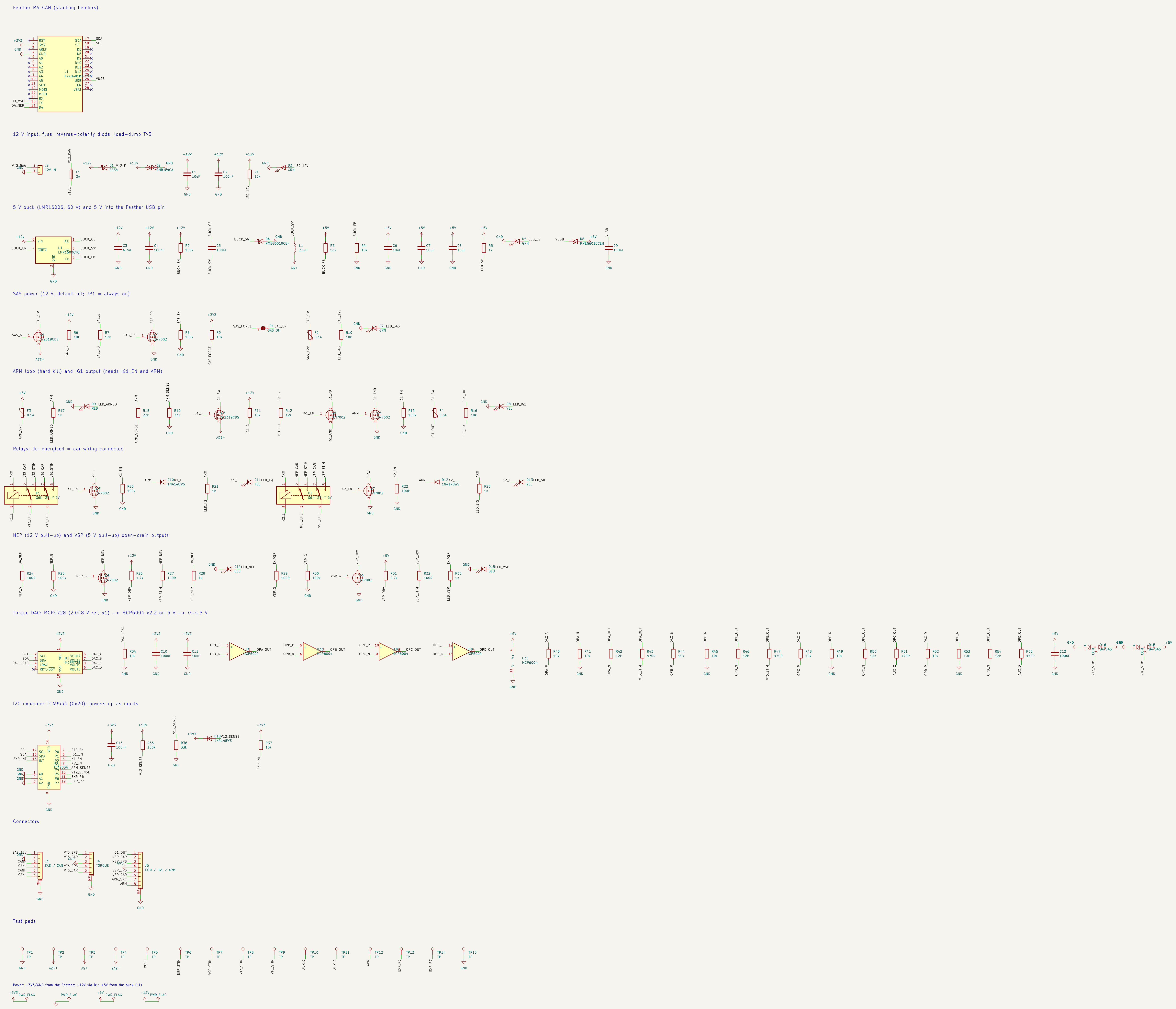
Bench stimulus FeatherWing: 12 V power, SAS power + CAN pass–through, EPS NEP/VSP/torque stimulus

Every EPS line goes through a relay whose NC contact is the car's own wire: no power, no ARM loop or no enable bit means the EPS sees the car exactly as without this board. ARM loop on J5 (pins 7–8): wire an NC e–stop or key switch. Faking VT3/VT6 makes the motor steer: wheels off the ground, hands clear. All enables come from U4 (TCA9534), which powers up with every pin an input; 100k pull–downs hold every driver off until firmware sets the bit.


Feather M4 CAN (stacking headers)
J1
Feather M4 CAN
TX_VSP
D4_NEP
VUSB
SDA
SCL
12 V input: fuse, reverse–polarity diode, load–dump TVS
J2 12V IN
F1 2A
D1 SS34
D2 SMBJ24CA
C1 10uF
C2 100nF
R1 10k
D3 GRN
5 V buck (LMR16006, 60 V) and 5 V into the Feather USB pin
U1 LMR16006YQ
C3 4.7uF
C4 100nF
R2 100k
C5 100nF
D4 PMEG6010CEH
L1 22uH
SAS power (12 V, default off; JP1 = always on)
Q1 SI2319CDS
Q2 2N7002
JP1 SAS ON
ARM loop (hard kill) and IG1 output (needs IG1_EN and ARM)
F3 0.1A
D9 RED
Q3 SI2319CDS
Relays: de–energised = car wiring connected
K1 G6K-2F-Y 5V
K2
D10 1N4148WS
D11 YEL
NEP (12 V pull–up) and VSP (5 V pull–up) open–drain outputs
D14 BLU
Torque DAC: MCP4728 (2.048 V ref, x1) –> MCP6004 x2.2 on 5 V –> 0–4.5 V
U2 MCP4728
U3 MCP6004
I2C expander TCA9534 (0x20): powers up as inputs
U4 TCA9534
D18 1N4148WS
Connectors
J3 SAS / CAN
J4 TORQUE
J5 ECM / IG1 / ARM
Test pads
TP1 TP
TP2 TP
TP3 TP
TP4 TP
TP5 TP
TP6 TP
TP7 TP
TP8 TP
TP9 TP
TP10 TP
TP11 TP
TP12 TP
TP13 TP
TP14 TP
TP15 TP
Power: +3V3/GND from the Feather; +12V via D1; +5V from the buck (L1)
PWR_FLAG

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Honda Insight EPS bench / in–car stimulus + 12 V power

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Title: Bench stimulus FeatherWing

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